S2-24_AIMLCZG530

Topic: Wordnet and Word Sense Disambiguation

NLP Assignment - 2

August 9, 2025 11.59pm TO August 24, 2025 11.59pm

Set 3

General Instructions:

1. Follow the instructions in each question carefully.
2. Each group is expected to submit jupyter notebook (.ipynb) with output for each cell.
3. Submissions using other python IDEs will not be considered for grading.
4. In case the link to dataset is not useful, same dataset can be downloaded from any online resource.
5. A clear explanation for each output obtained is mandatory.
6. Contribution of each member of the group must be mentioned.
7. Justification of the output obtained for all the tasks is mandatory.
8. Clearly mention the assumptions made if necessary.

Link to the Dataset:

The following data is intended for advancing financial sentiment analysis research

https://drive.google.com/file/d/15Wmi3GyCeEX6w8N528S40OYCnRhInH5W/view?usp=sharing

Description of Data:

The data set is intended for advancing financial sentiment analysis research

1. (5 Marks) (Mention suitable assumptions if necessary.)
    a. Download the dataset and Create a dataframe named as **fsa,** then check the head, info, and describe methods on created dataframe **fsa**. **(0.5 Mark)**
    b. Perform pre-processing steps like Removing Punctuations, Numbers, and Special Characters, Stop Words in dataset. **(0.5 Mark)**
    c. Normalize Sentence column by using Stemming or Lemmatization. **(0.5 Mark)**
    d. Demonstrate use of Sentiwordnet to calculate the senti_score of each of the reviews. (3.5 Marks)

2. Take some sample reviews from the dataset and demonstrate the following: (5 Marks)
    a. Synsets
    b. Synonyms and Antonyms
    c. Hyponym and Hypernym
    d. Wordnet path similarity
    e. Word sense disambiguation

(Mention suitable assumptions if necessary.)